Network Traffic Monitoring & Intrusion Detection using Zeek and Elastic SIEM

1. Project Overview

This project focuses on **network-level monitoring and intrusion detection** using **Zeek** as a network traffic analyzer and **Elastic SIEM** for centralized log analysis and visualization.

The goal is to:

- Capture live network traffic
- Analyze connection, DNS, and anomalous behavior
- Correlate network events in a SIEM platform
- Demonstrate SOC-style investigation without signature-based IDS

2. Lab Architecture

Components Used

- **Windows VM** – Traffic generator (browsing, ping, background services)
- **Ubuntu Server VM** – Zeek + Filebeat
- **Elastic Stack (Docker)** – Elasticsearch & Kibana

Traffic Flow

Windows VM → Network Traffic
↓
Ubuntu (Zeek)
↓
Filebeat
↓
Elasticsearch → Kibana (SIEM)

3. Tools & Technologies

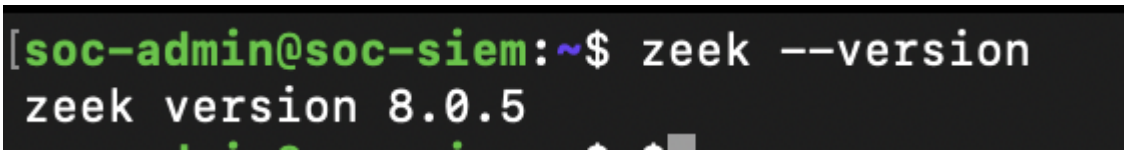
- Zeek (Network Security Monitor)
- Filebeat (Log Shipper)
- Elasticsearch
- Kibana (Discover / SIEM views)
- Ubuntu Server 22.04
- Windows VM

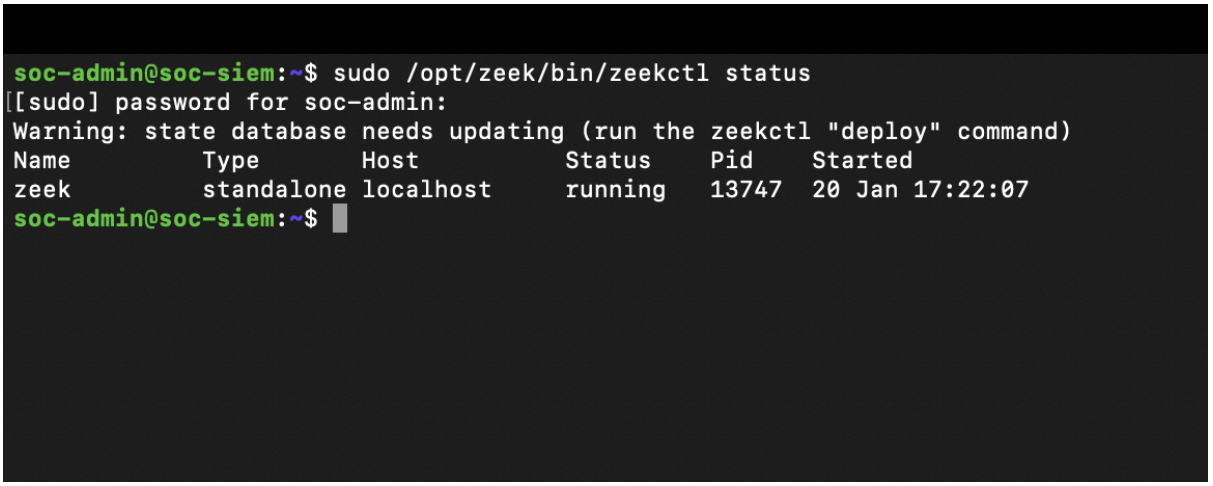
4. Zeek Installation & Service Verification

Zeek was installed on the Ubuntu Server and configured in **standalone mode** to monitor the active network interface.

Verification Commands

```
zeek --version
sudo /opt/zeek/bin/zeekctl status
```





5. Zeek Log Generation Verification

Zeek continuously generates logs representing different network behaviors.

Verified Log Directory

```
sudo ls /opt/zeek/logs/current
```

Observed Logs:

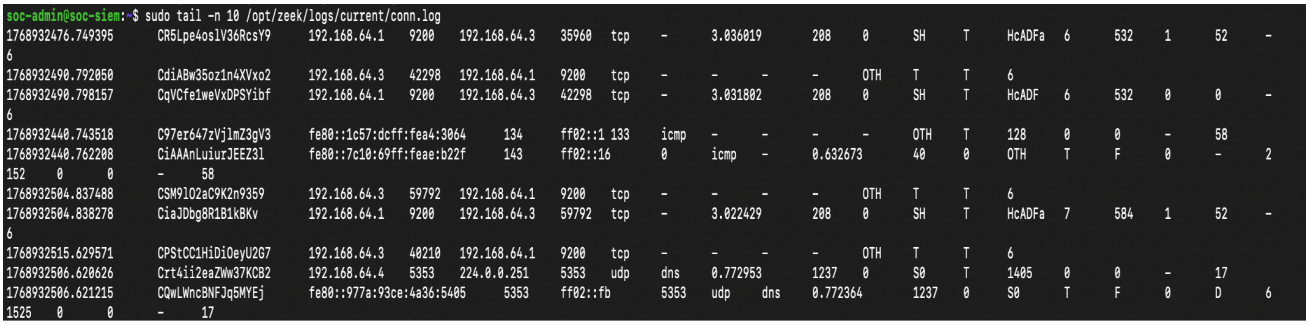
- `conn.log`
- `dns.log`
- `weird.log`
- `notice.log`
- `stats.log`



6. Network Connection Monitoring (conn.log)

The `conn.log` records **every network connection** including:

- Source IP / Port
- Destination IP / Port
- Protocol



7. DNS Activity Monitoring (dns.log)

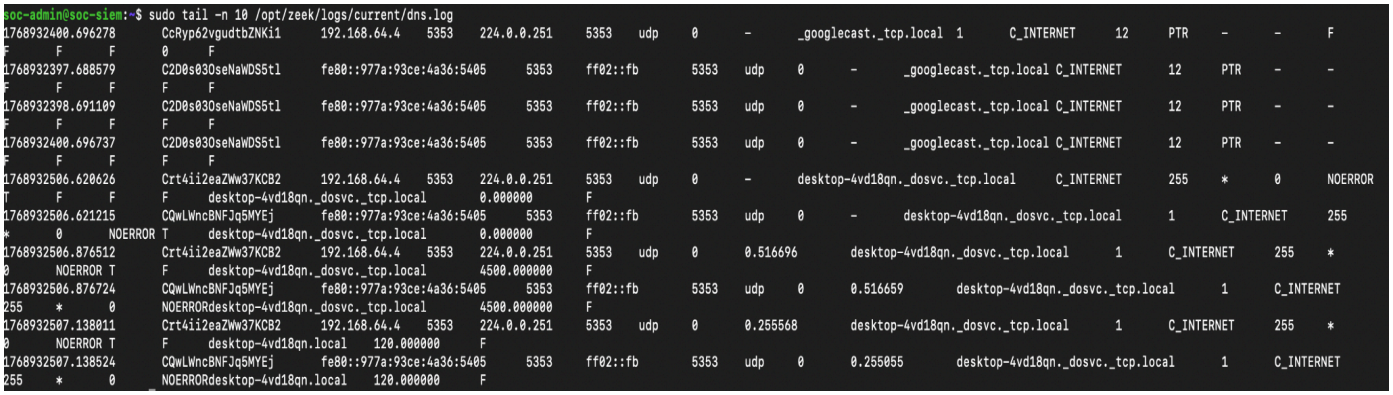
The `dns.log` records **all DNS queries and responses**, including multicast DNS traffic.

Command Used

```
sudo tail -n 10 /opt/zeek/logs/current/dns.log
```

Observations

- Local hostname resolution
- Multicast DNS (`.local`)
- Service discovery traffic



8. Anomaly & Intrusion Detection (weird.log)

Zeek flags protocol-level anomalies in `weird.log`.
These are **NOT signatures**, but behavioral deviations.

Command Used

```
sudo tail -n 10 /opt/zeek/logs/current/weird.log
```

Observed Anomalies

- `data_before_established`
- `active_connection_reuse`

- `inappropriate_FIN`

These indicate:

- TCP misuse
- Abnormal session handling
- Potential reconnaissance or malformed traffic

```
soc-admin@soc-siem:~$ sudo tail -n 10 /opt/zeek/logs/current/weird.log
1768932637.271478       CGgvbz20389lsbkUv3      192.168.64.1    9200    192.168.64.3    35262   inappropriate_FIN       -       F       zeek    TCP
1768932646.855153       Cjr0nJ3zr1zj9X0AG9      192.168.64.3    47572   192.168.64.1    9200    active_connection_reuse -       F       zeek    TCP
1768932646.868703       CEi1SQ30pbBoezSmIe      192.168.64.1    9200    192.168.64.3    47572   data_before_established -       F       zeek    TCP
1768932649.878215       CEi1SQ30pbBoezSmIe      192.168.64.1    9200    192.168.64.3    47572   inappropriate_FIN       -       F       zeek    TCP
1768932658.343884       CTooyv43zk5vl1Brv8      192.168.64.3    51962   192.168.64.1    9200    active_connection_reuse -       F       zeek    TCP
1768932658.360049       CdE0SYpxk31mSZCUb       192.168.64.1    9200    192.168.64.3    51962   data_before_established -       F       zeek    TCP
1768932661.370859       CdE0SYpxk31mSZCUb       192.168.64.1    9200    192.168.64.3    51962   inappropriate_FIN       -       F       zeek    TCP
1768932672.361863       CatW1H2ZzAy9hDIwvf      192.168.64.3    51416   192.168.64.1    9200    active_connection_reuse -       F       zeek    TCP
1768932672.371294       Cl0MTO2IQnjF0XpOhg      192.168.64.1    9200    192.168.64.3    51416   data_before_established -       F       zeek    TCP
1768932675.378708       Cl0MTO2IQnjF0XpOhg      192.168.64.1    9200    192.168.64.3    51416   inappropriate_FIN       -       F       zeek    TCP
```

9. Filebeat Installation & Status Verification

Filebeat was installed and configured to forward Zeek logs from:

`/opt/zeek/logs/current/*.log`

to Elasticsearch.

Verification Commands

```
filebeat version
sudo systemctl status filebeat
```

```
soc-admin@soc-siem:~$ filebeat version
sudo systemctl status filebeat
filebeat version 8.19.10 (amd64), libbeat 8.19.10 [9c2978ab88f543f7ef768e388216c0725ab9c456 built 2026-01-08 19:03:28 +0000 UTC] (FIPS-distribution: false)
[sudo] password for soc-admin:
● filebeat.service - Filebeat sends log files to Logstash or directly to Elasti>
     Loaded: loaded (/lib/systemd/system/filebeat.service; disabled; vendor pre>
     Active: active (running) since Tue 2026-01-20 11:36:18 UTC; 7h ago
       Docs: https://www.elastic.co/beats/filebeat
   Main PID: 8114 (filebeat)
      Tasks: 12 (limit: 6962)
     Memory: 41.8M
        CPU: 3min 48.452s
     CGroup: /system.slice/filebeat.service
             └─8114 /usr/share/filebeat/bin/filebeat --environment systemd -c />
```

10. Elasticsearch Log Ingestion Verification

Index Verification

Elasticsearch indices confirm successful ingestion of Zeek logs.

Command (Kibana Dev Tools)

```
GET _cat/indices?v
```

```
health status index                                    uuid                   pri
  rep docs.count docs.deleted store.size pri.store.size dataset.size
yellow open   .ds-winlogbeat-9.2.3-2026.01.20-000001 S3U6vXSBRNSWETpfvzzJkg   1
    1        943            0      1.9mb          1.9mb        1.9mb
yellow open   .ds-filebeat-8.19.10-2026.01.20-000001 UbM6p8_PTdWJ368ETgjjEQ   1
    1       5257            0      1.5mb          1.5mb        1.5mb
```

11. Kibana SIEM Analysis (Core Evidence)

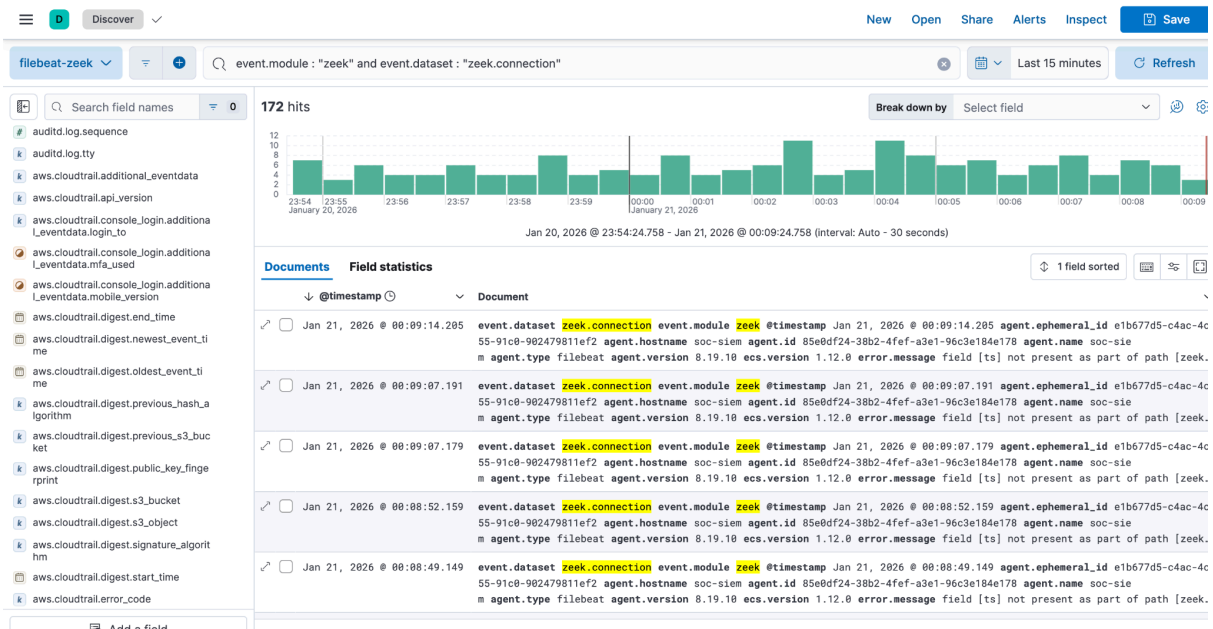
11.1 Network Connections (Zeek Connection Logs)

KQL Query

```
event.module:"zeek" and event.dataset:"zeek.connection"
```

Purpose

- Visualize all observed network sessions
- Identify frequent connections and timing patterns



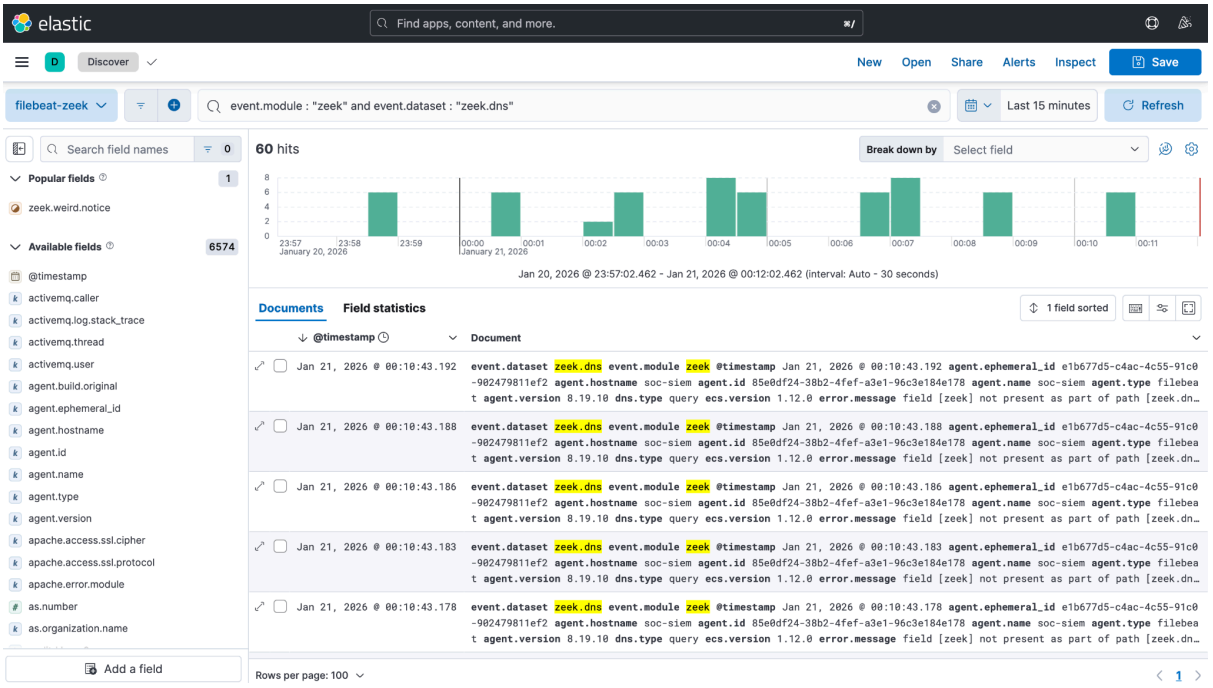
11.2 DNS Monitoring

KQL Query

```
event.module:"zeek" and event.dataset:"zeek.dns"
```

Purpose

- Monitor domain lookups
- Detect unusual or excessive DNS queries



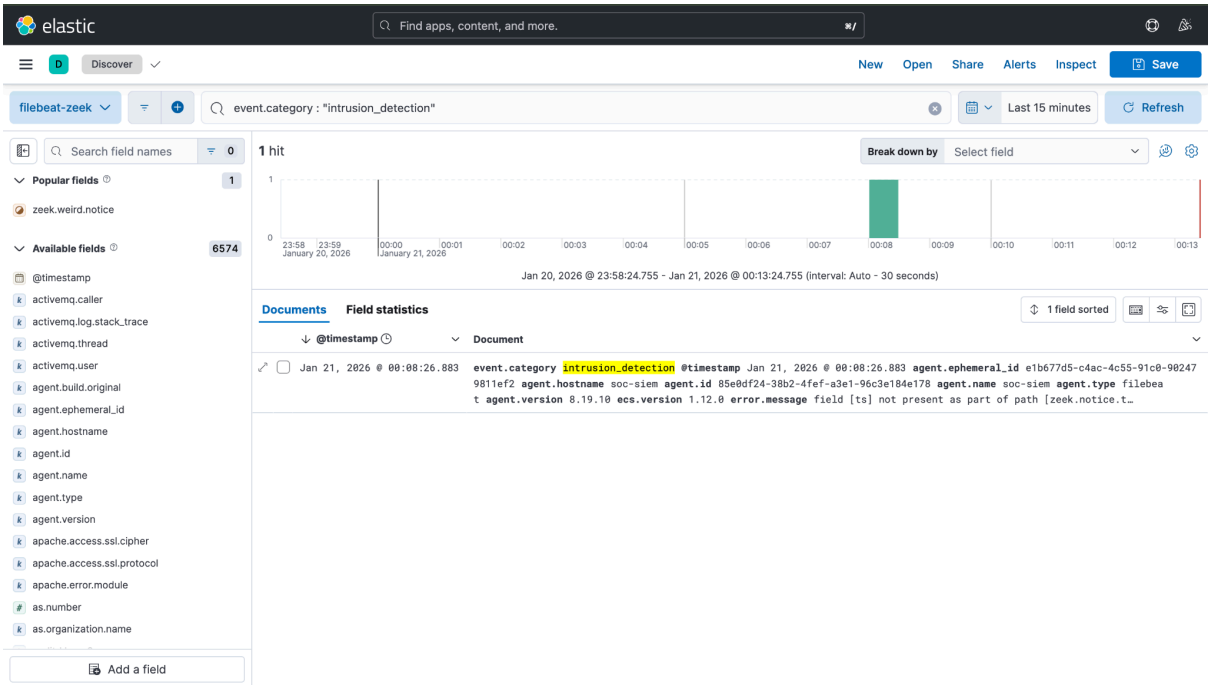
11.3 Intrusion Detection Events

KQL Query

```
event.category:"intrusion_detection"
```

Purpose

- Highlight suspicious protocol behaviors
- Surface Zeek weird/notice events



12. Traffic Scenarios Demonstrated

Scenario 1: Normal User Activity

- Website browsing
- DNS resolution
- TCP connections to services

Scenario 2: Anomalous Network Behavior

- TCP state violations
- Improper FIN packets
- Reused active connections

These were **detected automatically by Zeek** and surfaced in Elastic SIEM.

13. SOC Analyst Value

This project demonstrates:

- Network-centric detection
- Behavioral analysis over signatures
- SIEM-based investigation workflow
- Real SOC-style traffic visibility

14. Conclusion

Project 3 successfully implements **network traffic monitoring and intrusion detection** using Zeek and Elastic SIEM.
It proves that meaningful security insights can be derived from **protocol behavior analysis** without relying on traditional IDS rules.